SAP-C02 Domain 2 — Design for New Solutions

Average question, answer template and traps for tasks 2.1–2.6 · built from the 241 Domain 2 questions in your trainer

The average Domain 2 question

> "A company is building **one new workload** (web app, API, pipeline, DR copy) with **EC2 / Lambda / ECS / RDS / DynamoDB / S3**. It must **scale with spikes / survive an AZ or Region loss / stay off the internet / never lose data / cost less**. Which solution meets these requirements with **LEAST overhead** (or **MOST cost-effectively**)?"

The average answer has 4 parts

- **Where:** the right building block per requirement (compute, data store, queue, edge).
- **What:** a managed service with native integration, no glue code.
- **How it survives:** multi-AZ by default; a second Region only for DR or global users.
- **How it's protected:** roles (not keys), Secrets Manager, endpoints + private DNS, WAF at the front door.

The wording tells you which answer wins

- **LEAST overhead** → managed / serverless, built-in features.
- **MOST cost-effective** → cheapest option that *just* meets the numbers; pay per use, Spot, reuse.
- **Highest availability / fastest failover** → multi-AZ + health-checked routing; Aurora global, CloudFront origin group.
- **"never lose" / "spikes"** → queue or stream between tiers (SQS + DLQ, Kinesis).
- **"no internet" / "private"** → VPC endpoints + private DNS, PrivateLink, Client VPN.
- **"prevent" / "must not"** → a real block (policy, SCP, DeletionPolicy, Block Public Access), never Config alone.
- **"over 15 min" / "waits for a human"** → Fargate or Batch / Step Functions task token.

The 5 workload checks

Check	Rule
What runs it?	Fit runtime limits: Lambda ≤15 min; Fargate or Batch longer; EC2 only for OS control, licences, special hardware.
What fails?	Every tier in 2+ AZs (ASG, NAT gateway per AZ, Multi-AZ DB). A Multi-AZ standby is **not** DR; DR needs data in another Region.
What's between the tiers?	Queue or stream if load spikes or nothing may be lost; EventBridge or SNS to fan out.
Who can reach it?	Roles, not keys; Secrets Manager rotation; private paths via endpoints; WAF / Shield / Cognito at the front door.
What does it cost idle?	Prefer paying when it runs: serverless, Spot, Athena, Intelligent-Tiering, gateway endpoint.

Reading routine

- **Last sentence first:** MOST/LEAST word, how many answers (48 of 241 ask for TWO or THREE).
- Underline: **the workload** · **hard numbers** (RTO/RPO, minutes, Mbps, requests/s) · **hard limits** (no internet, fixed IP, UDP, no code change, not Aurora).
- Cross out: options breaking a hard limit, self-managed EC2 where a managed service exists, custom glue, the facts below.

Wrong almost every time: self-managed EC2 for what a managed service does (58 wrong vs 30 right) · Lambda as glue, polling or for jobs over 15 min (74 wrong vs 47 right; fine as the serverless compute piece) · one NAT gateway or one AZ "for HA" · RDS Multi-AZ standby as cross-Region DR · stack policy to stop deletion · ALB with an Elastic IP or UDP · CloudFront for UDP or unusual HTTP methods · HTTP API usage plans or caching · Parameter Store "rotation" · S3 interface endpoint when cost matters · Config "preventing" anything · backups every 30 seconds · Route 53 failover without health checks / evaluate target health.

2.1 Deployment strategy

> **Q:** Release or run [app] with [no downtime / canary / patched instances / runs over 15 min / fixed outbound IP].
> **A:** The service's **own deployment feature**, or the compute that **fits the runtime**.

Need	Answer
Lambda canary	Alias with weighted routing (CodeDeploy)
ECS / Beanstalk blue/green	ECS blue/green + ALB · Beanstalk **swap URLs**
Patched Auto Scaling group	New AMI → launch template → **instance refresh**
Keep data if stack deleted	**DeletionPolicy** Retain / Snapshot
Job over 15 min / infrequent	**Fargate** started by EventBridge

Need	Answer
Per-customer API quota	**REST API usage plan + API key**
Fixed outbound IP	**NAT gateway + Elastic IP**
Lowest latency between nodes	**Cluster placement group**
Instance missing in Systems Manager	SSM Agent running + IAM role
Wait for outside approval	Step Functions **task token**

Checks: runtime limit? · native deployment feature? · fixed IP or private path?

2.2 Business continuity

> **Q:** Survive loss of [AZ / Region] with RPO X / RTO Y, at [lowest cost / fastest failover].
> **A: Replicate the data natively** + **build the second Region with IaC** + **switch with health-checked routing**.

Need	Answer
RPO ~1 s, RTO minutes	**Aurora global database**
Cheaper relational DR	Cross-Region read replica
NoSQL in both Regions	**DynamoDB global tables**
Reconnect without restart / faster failover	**RDS Proxy**
EBS snapshots to other Regions	**Data Lifecycle Manager**

Need	Answer
Switch Regions	Route 53 **failover + evaluate target health**, short TTL
Fastest switch for web content	**CloudFront origin group**
Back up everything + alerts	**AWS Backup**, cross-Region copy, SNS
On-premises servers	**Elastic Disaster Recovery** (+ drills)

Checks: RTO/RPO → cheapest pattern that fits · is the **data** in the other Region? · does routing check health?

2.3 Security controls

Q: [Who] must reach [what] with [least privilege / no internet / rotated secrets / MFA / encryption].
A: Role or policy for access + **Secrets Manager** for credentials + **endpoints** for private paths + **customer managed key** when a team controls the key.

Need	Answer
Outside auditors	Role + **ExternalId**, read-only
Minimal policies for Lambda	**Access Analyzer** from CloudTrail activity
Rotate DB passwords	**Secrets Manager rotation** (or IAM DB auth)
API only from the VPC	**Private API + interface endpoint** + resource policy
Endpoint resolves to public IPs	Turn on **private DNS**

Need	Answer
Python packages, no internet	**CodeArtifact + endpoint**
Remote developers into the VPC	**Client VPN** (+ AD Connector for MFA)
Per-user S3 folders	`${aws:PrincipalTag/userName}`
Public bucket, fix it now	**Block Public Access**
Container / Lambda CVEs	**Inspector / ECR enhanced scanning**

Checks: both sides allow? · temporary credentials? · traffic stays private?

2.4 Reliability

Q: Handle spikes / failures / global users without losing requests.
A: Queue between tiers + **every tier in multiple AZs** + **health-checked routing** for global users.

Need	Answer
Spiky writes, nothing lost	**SQS → Lambda** + **dead-letter queue**
Errors to a review queue	Lambda **on-failure destination** / DLQ
One AZ fails	ASG across AZs, **NAT gateway per AZ**, Multi-AZ DB
Closest Region + failover	**Latency routing + evaluate target health**
UDP / static IPs / apex / custom methods	**Global Accelerator**

Need	Answer
Only CloudFront may reach the ALB	**CloudFront prefix list** or secret header
Block a country, log blocks	**WAF geo rule**
AWS → on-prem DNS	**Outbound endpoint + rule**
On-prem → AWS DNS	**Inbound endpoint**
Overlapping CIDRs to one app	**PrivateLink**

Checks: what happens at a spike? · single AZ anywhere? · routing checks health?

2.5 Performance

Q: [Stream / query / file workload] is too slow or loses data at peak.
A: Buffer the stream, **shape the data** for queries, **pick storage by protocol and access pattern**.

Need	Answer
Burst sensor data	**Kinesis Data Streams** (Lambda / Fargate consumers)
Devices with certificates	**IoT Core** (+ pre-provisioning hook)
Athena keeps getting slower	**Partitions + Parquet + bigger files**
HPC at scale	**One AZ + EFA + FSx for Lustre**

Need	Answer
Shared POSIX across AZs	**EFS**
Windows with ACLs	**FSx for Windows**
Unknown access pattern	**S3 Intelligent-Tiering**
Approved launches with tags	**Service Catalog**

Checks: buffer or stream? · scanning too much data? · right protocol?

2.6 Cost

Q: Same result, MOST cost-effectively.
A: Pay only when running, **reuse what already exists**, **match each commitment to its service**.

Need	Answer
Fresh S3 data on EC2	**Mountpoint for S3**
Old search data	OpenSearch **UltraWarm** → S3
Retry-safe batch	**Batch on Spot**
Short SQL jobs	**Athena** instead of a running cluster
Data that expires	**DynamoDB TTL** / S3 Lifecycle

Need	Answer
Many ML models	**Multi-model endpoints**
S3 traffic through NAT	**Gateway endpoint**
Bulk transfer, Direct Connect exists	Use the **existing Direct Connect**
Mixed services	**One commitment per service**
ECS cost per team	**Split cost allocation data**

Checks: is anything idle? · does it already exist? · does the commitment cover this service?